



# AboutBits

# Beyond Prompting How We Use AI in Software Development

# About Us

- Custom software development agency
- Based in Klausen
- Team
  - 14 developers
  - 2 designers
- We are constantly challenging how we work, and AI will have a huge impact on it.

# Intro



Man... it's exhausting.  
Models, Tools,  
Patterns,...



# Our philosophy

## Experiment and share

- It's impossible to keep up as an individual
- Every developer and designer is invited to test different tools and patterns
- We buy commercial licenses for tools requested by our employees
- Everybody is encouraged to apply an AI first approach and optimize their workflows accordingly
- Most importantly share the learnings with the team



**Andreas Hufler** 😎 21:55 Uhr

Hier ein Block für die `claude.md` oder Junie `guidelines.md` der ganz hilfreich ist:

```
# General Guidelines
```

```
## Challenge Decisions
```

- When I present a plan, approach, or assumption, actively look for flaws before helping execute it.
- Ask "Is this actually the right problem to solve?" before diving into solutions.
- If you disagree with my framing or approach, say so directly – don't just comply silently.
- Point out when I'm optimizing the wrong thing (e.g., "This is fast, but you haven't verified it works correctly").
- Flag when a decision seems driven by sunk cost, habit, or bias rather than reasoning.

# Goal of the talk

## Sharing our experience

- Share our experience working with agentic coding tools taking Claude Code as reference
- Showcase a few experiments and ideas that we play around with
- Hopefully have a great discussion and learn from your experiences

# Claude Code

## Agentic coding

- Agentic coding tool by Anthropic
- Uses Anthropic LLMs, such as Opus 4.7, to reason about tasks and code bases
- It uses different tools to act on your code base
  - Read, Edit, Create, Rename, Organize files and folders
  - Search and explore
  - Run shell commands
  - Search the web
  - Interact with external services
- Download CLI and start coding

```
[→ tech-talks claude ]
Claude Code v2.1.150
Sonnet 4.6 · API Usage Billing
~/development/playground/tech-talks

> Try "fix lint errors"

? for shortcuts · ← for agents      2 MCP servers need auth · /mcp
```

# CLAUDE.md

## Persistent context

### Project Guidelines

#### Code Style

- Follow existing patterns in the codebase
- Use meaningful variable names
- Keep methods focused and concise

#### Commands

```
# Install dependencies
npm ci

# Dev servers (admin on :3000, tenant on :3001)
npm run admin:dev
npm run tenant:dev

# Build
npm run admin:build
npm run tenant:build

# Lint
npm run lint
npm run lint:fix

# Type check
npm run typecheck
```

- Claude has access to all files in your directory and subdirectories, plus files elsewhere with your permission.
- Without any further context, it is up to Claude to figure out by itself all about your project.
- **Claude.md** is a persistent context, which is loaded into every session.
- Keep it short (**~200 lines**) and concise
- What should go into CLAUDE.md
  - General standards and guidelines
  - Important commands (build, type check, lint, test)
  - Architecture of the project
- Use ``.claude/rules`` if you want context which should only be loaded if the agent works on certain files.

# CLAUDE.md

## General purpose

- Four rules from Andrej Karpathy, that went viral on X in January 2026  
<https://x.com/karpathy/status/2015883857489522876>
- <https://github.com/multica-ai/andrej-karpathy-skills>
  - Think Before Coding
  - Simplicity First
  - Surgical Changes
  - Goal-Driven Execution
- Global CLAUDE.md (~./claude/CLAUDE.md)

# Skills

## Reusable workflows

- Skills extend agentic coding tools with additional capabilities.
- Very useful for repetitive tasks, that you can define as a certain procedure.
  - For example a certain refactoring pattern that you want to do on different parts of your app.
- Claude loads only the names of the skills into every new context and during session it loads skills on demand.
- Skills can also be called as commands (/skill-name). Like that you don't waste unnecessary tokens.
- Skills are in it's simplest form a Markdown file, but it can also contain scripts.

# Skills

## Project specific

- Located in your repo ``.claude/skills``
- Some examples
  - Migrate Hibernate / Spring Data JPA to jOOQ
  - Add typed ids to legacy projects
  - Replace testing setups with new pattern
  - Generate CRUD code
- Things that worked well for us
  - Keep the scope small
  - Provide short and concise description of the skill
  - Add code examples or reference code with best practices
  - Constantly update skills
- Things we currently consider
  - Company wide skills (e.g. Naming conventions for liquibase migrations)

# Example: Migrate Model ID from Long to Typed ID

|             |                                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name        | migrate-id-to-typed                                                                                                                                                                                                                                     |
| description | Use this skill when you need to migrate a model's ID field from Long to a typed ID (e.g., <code>ModelName.ID</code> ), when the user asks to "migrate IDs to typed IDs", "add typed ID to [Model]", or "use <code>ModelName.ID</code> instead of Long". |

## Migrate Model ID from Long to Typed ID

This skill helps you migrate a domain entity's ID field from `Long` to a typed ID wrapper (e.g., `Benefit.ID`) following the pattern established in the codebase. This provides better type safety and prevents accidental ID mixing between different entity types.

### Quick Reference

1. Add `ID` record to the model class and replace `Long id` with `ID id`
2. Update model interfaces: `Identified<ClassName.ID>`, `ChangeAware`, and `BaseEntity<ClassName.ID>`. The `BaseEntity<ClassName.ID>` interface should only be updated, do not add it if the original code has none.
3. Update JPA repository to use `ClassName.ID` instead of `Long` in all methods
4. Update `ManageClassName` and `QueryClassName` to use `ClassName.ID` instead of `Long`

# Skills

## Public skills

- Frameworks, libraries, CLIs, etc are nowadays shipping skills which you can add to your project.
  - Google Workspaces released their new CLI with corresponding skills.
  - We used Payload CMS skills successfully

### Installation

```
npx skills add payloadcms/skills
```

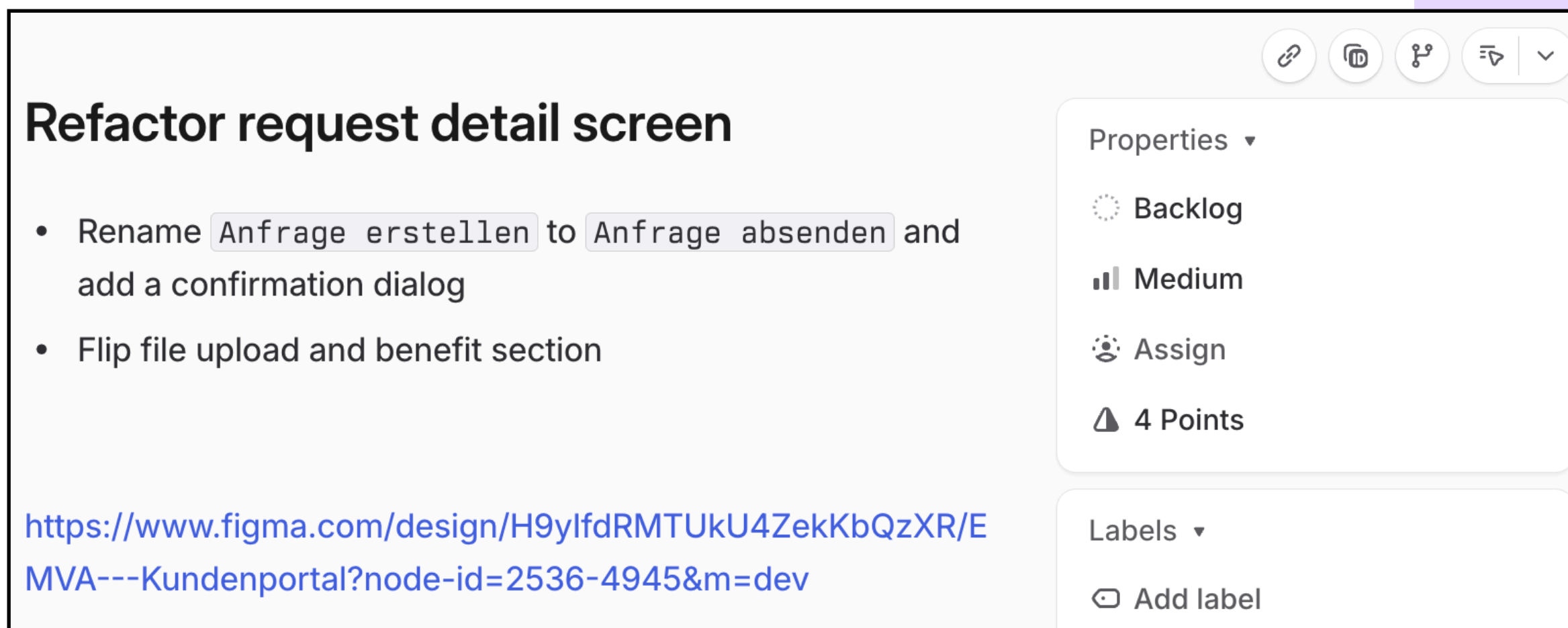


- General purpose skills
  - **grill-me** from Matt Pocock  
<https://github.com/mattpocock/skills/tree/main/skills/productivity/grill-me>
  - **skill-creator** from Anthropic
- <https://www.skills.sh/>

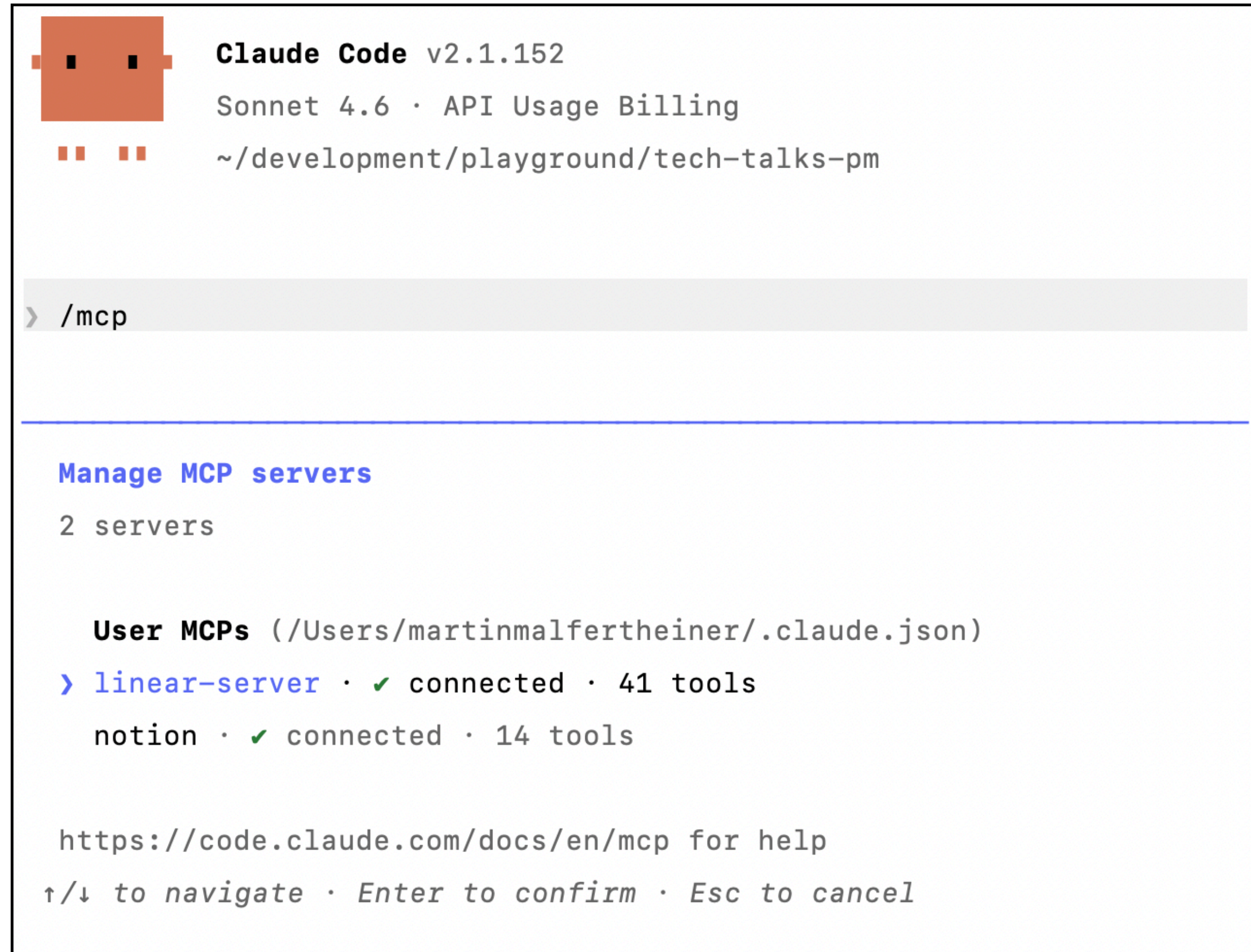
# MCP

## Model Context Protocol

- Protocol to connect to external tools and data sources
- MCP servers give Claude Code access to your tools, databases, and APIs
- Claude determines during runtime when to use an MCP service (was not always like that)
- „Implement linear issue AB-123 and create a PR“
  - Claude will use the **Linear MCP** server to fetch the ticket details
  - It will read the body and see the Figma link
  - It will then use the **Figma MCP** server to fetch meta data and screenshots about the design
  - It will use **Github MCP** to finally create a PR with Branch name from **Linear** for the feature.



# MCP: Open terminal



The image shows a terminal window from Claude Code v2.1.152. The window has a title bar with the Claude logo and version. Below the title bar, it shows the current directory as ~/development/playground/tech-talks-pm. A search bar contains the text '/mcp'. The main content area displays the 'Manage MCP servers' menu, showing 2 servers: 'linear-server' (41 tools) and 'notion' (14 tools). At the bottom, there is a help link and navigation instructions.

```
Claude Code v2.1.152
Sonnet 4.6 · API Usage Billing
~/development/playground/tech-talks-pm

> /mcp

Manage MCP servers
2 servers

User MCPs (/Users/martinmalfertheiner/.claude.json)
> linear-server · ✓ connected · 41 tools
notion · ✓ connected · 14 tools

https://code.claude.com/docs/en/mcp for help
↑/↓ to navigate · Enter to confirm · Esc to cancel
```

# Self verification

## Invest in testing infrastructure

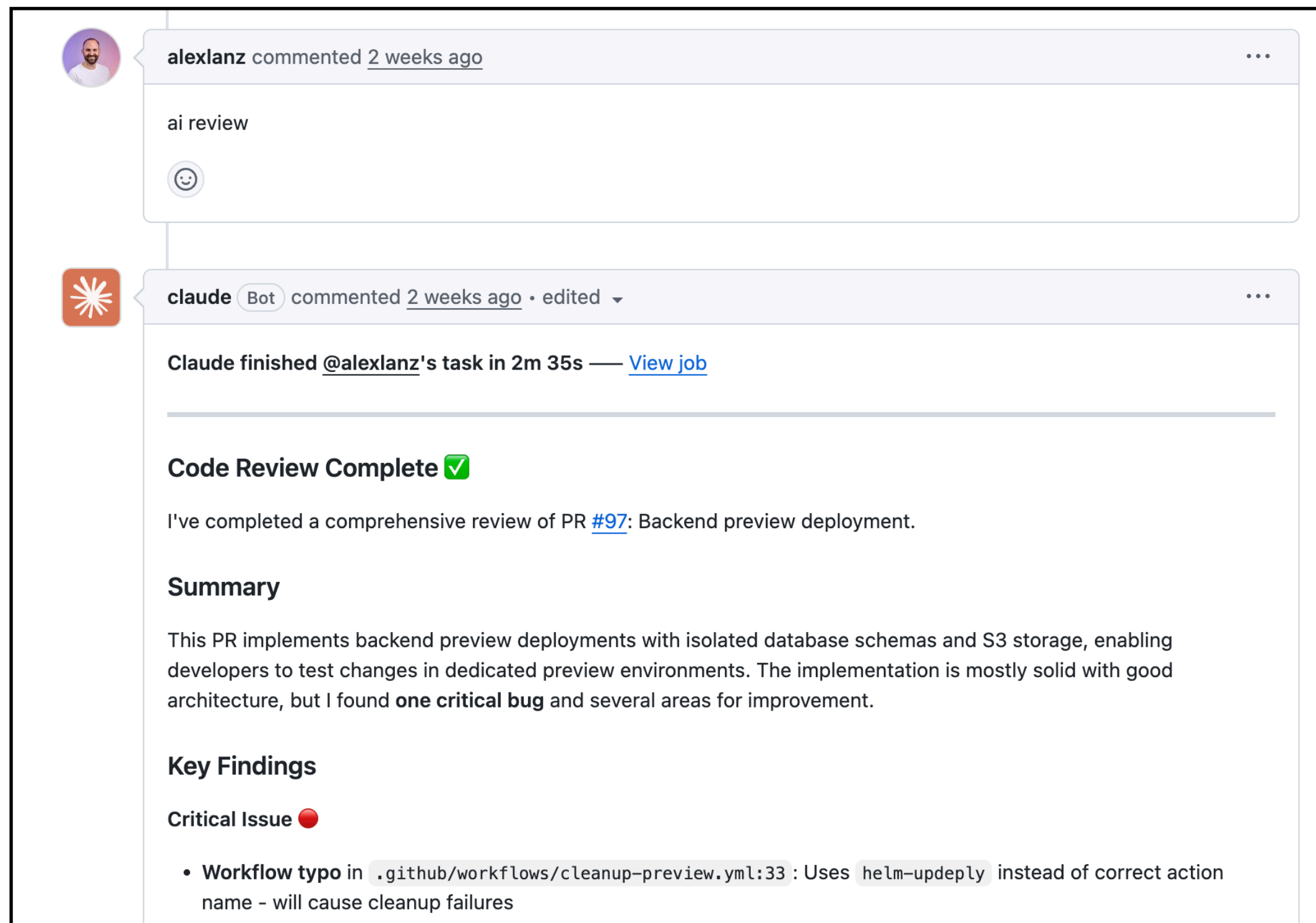
- Self verification
  - The agent needs a way to verify its goal
  - Static code analysis, test infrastructure, test scenario setups, architecture tests
- Custom effort for specific tasks
  - We have a code generator that is based on handlebars
  - A colleague wants to migrate to Typescript string templates
  - Before starting the task he elaborated a script that gives the agent a way to verify the output
  - <https://github.com/aboutbits/finstral-composer-api/pull/691/changes#diff-b8a0df04774ba3205b1298f8930a8c07ba8b7c19bf3ed07b3167adda090f3b02>

# Experiments & Inspiration



# Claude Github Action

## Code review



- We use it for code reviews
  - Add a comment „ai review“
  - Github Action runs code review
  - Picks up context from your repository
- We are not 100% happy with it
  - It is quite noisy, which adds work to check the comments from AI
  - Developers still needs to do a proper review
  - No influence on the default prompt
- Current experiments
  - Self review locally by the developer using a local skill
  - Local running agent that listens to Github comments and does a code review

# Developer tooling

## Vibe coding

- Building personal tools and scripts is very cheap with AI
- One of my colleagues build a personal assistant that manages his daily work
  - What do I do the whole day and how can I make my life easier?
  - Build a dashboard that integrates with Github, Linear, Calendar, Time tracking app and more.

The screenshot shows a personal dashboard with the following sections:

- Tasks**: 53 active. Includes tabs for Tasks, Workunits, STT, Riker, Questions, and a filter. A search bar says "Ask a question or give a command... (⌘↵ to send)".
- TODAY'S PLAN**: Includes a "Re-plan" button. Shows a timeline with "4h in meetings".
  - 09:00 – 09:15: Standup, 30m booked, ✓ done.
  - 13:30 – 17:00: Photoshooting, 3h 30m, ✗.

1h 13m remaining · 1h 4m planned

1 ~64m P1

✗ Add OfferPosition persistence layer  
blocking — must do today (partial)

blocking — didn't fit today

Review Review: FINC-1092 BE: Tenant: ... ↗
- TOMORROW'S PLAN**: Empty section.
- CHORE**: quick task from the backlog.
  - project-riker**: P5 Push commits in project-riker (~10m, 4w ago). Includes a note about unpushed commits and buttons for "task:commit-push-project-riker", "project-riker", "Create Linear Ticket", and "Link Tyme".
  - FINC-1356**: P1 Add OfferPosition persistence layer (~1920m, 2w ago). Includes a "Blocking" status and a note about blocking downstream work.
- TIME TRACKING**: Includes a "+ Start tracking" button.
  - TODAY 2026-05-27**: 3h 17m.
    - Composer - Development > Milestone 6.4 > ... 10:24 – 12:10 1h 46m BE: Change plop gen...
    - Composer - Development > Milestone 6.1 > B... 09:49 – 10:24 35m Review Review: FINC-1...
    - Transcon > Support Christian Gapp 09:30 – 09:49 19m Review Review: Tra 28...
    - Team > 02. Standup Meeting 09:00 – 09:30 30m Standup [gcal\_\_6cqka...
    - Team > 04. Frontend Development 08:53 – 09:00 7m Review Review: RFC out...
  - LAST DAY 2026-05-26**: 7h 13m.

# Developer tooling

## Vibe coding

- IntelliJ plugin to support voice interaction with AI assistant

Type your task, use @ to add files or / for commands

+ ✨ Claude Agent ▾ Mode: Auto ▾ Model: Sonnet 4.6 (1M context) ▾ Effort: High ▾ 🗑️ 🎤 ✨

# Linear automation with Github Copilot

## Asynchronous coding agent

- In Linear it is possible to assign a ticket to Github Copilot
- Github Copilot will create a PR, work on the feature and when done request a review
- Github provides also interface to interact with the agent
- Advantages
  - Fully asynchronous
  - Well integrated with Linear
  - Transparent as it adds the assigned user as Co-Author
- Disadvantages
  - Closed system, which makes it hard to optimize prompts or to add external services like Figma MCP or Notion
  - Manual intervention is quite cumbersome

# Linear automation with Github Copilot

AboutBits

Pulse

Inbox

My issues

Reviews3

Drafts1

Workspace

Your teams

TRA-270 BE: periodic special-p

AB-451 Replace Nginx Ingress

Infrastructure > AB-451 Replace Nginx Ingress

Replace Nginx Ingress

The Nginx Ingress is deprecated and we need to think of an alternative:  
<https://kubernetes.io/blog/2025/11/11/ingress-nginx-retirement/>  
That's why I would like to collect different opinions and solutions in this ticket.  
⇒ We decided to go with Envoy Proxy

+ Add sub-issues

Resources

feat(AB-451): Replace Nginx Ingress with Envoy Gateway (Gateway API) #283 Closed

Activity

Alex Lanz created the issue · 4mo ago

Alex Lanz · 4mo ago  
@Simon Planinschek @Thomas Sapelza I would like to invite you to write some proposals here for alternative solutions.

Properties

In Progress

High

Alex Lanz

GitHub Copilot

Set estimate

Labels

Add label

Project

Infrastructure

# Linear automation with Github Copilot

aboutbits / infrastructure

Type / to search

>\_

<> Code

Pull requests

Agents

Actions

Security and quality

Insights

Settings

Created by me

+

No active sessions

Collecting solutions for Nginx Ingress replacement

...

Open in VS Code

You created the session 3 weeks ago · Claude Sonnet 4.6 · 1 session · 1 premium request

You started a task from [Linear](#)

Collecting solutions for Nginx Ingress replacement >

✓ 12m 22s

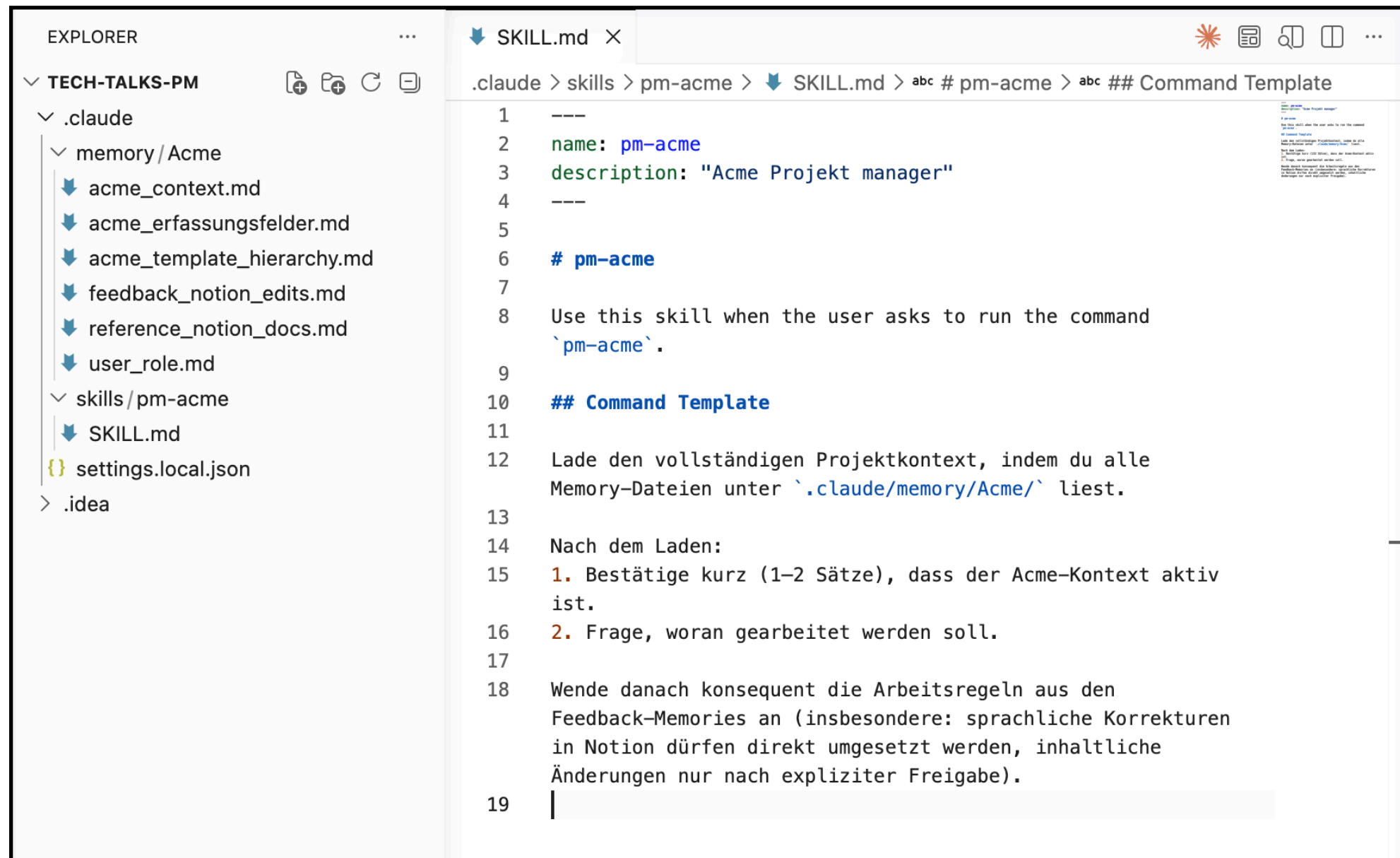
...

24 | Tech Talks South Tyrol

# Product management

## Writing documentation

- Simple folder
  - Memory about project specificities
  - Explanation of certain terms and concepts
  - Guidance about my personal writing style
- Skill that starts the conversation and loads the context
- MCP connection to Notion to read existing docs and update/add new docs



The screenshot shows a code editor interface. On the left is the 'EXPLORER' sidebar with a tree view of a project named 'TECH-TALKS-PM'. The tree structure is as follows:

- TECH-TALKS-PM
  - .claude
    - memory / Acme
      - acme\_context.md
      - acme\_erfassungsfelder.md
      - acme\_template\_hierarchy.md
      - feedback\_notion\_edits.md
      - reference\_notion\_docs.md
      - user\_role.md
    - skills / pm-acme
      - SKILL.md
    - settings.local.json
  - .idea

The main editor area shows the content of 'SKILL.md' with the following text:

```
1  ---
2  name: pm-acme
3  description: "Acme Projekt manager"
4  ---
5
6  # pm-acme
7
8  Use this skill when the user asks to run the command
9  `pm-acme`.
10
11 ## Command Template
12
13 Lade den vollständigen Projektkontext, indem du alle
14 Memory-Dateien unter `.claude/memory/Acme/` liest.
15
16 Nach dem Laden:
17 1. Bestätige kurz (1-2 Sätze), dass der Acme-Kontext aktiv
18   ist.
19 2. Frage, woran gearbeitet werden soll.
20
21 Wende danach konsequent die Arbeitsregeln aus den
22 Feedback-Memories an (insbesondere: sprachliche Korrekturen
23 in Notion dürfen direkt umgesetzt werden, inhaltliche
24 Änderungen nur nach expliziter Freigabe).
```

# Design

## Claude Design / Figma Make

- Designers experiment with Claude Design and Figma Make
  - Elaborate prototypes by non designers
  - Quickly explore different interactions
  - Build dev ready screens
- Based on actual design system components and rules
- Figma agents are in beta
  - Maintain design systems
  - Adjust tokens
  - Check conventions
- Next step: Connect Github repositories with actual components

# Closing remarks



# Closing remarks

## Experiment and share

- AI first mindset
  - How can AI help me to be more productive?
  - Challenge existing workflows: Are they still relevant in the age of AI?
- Share your learnings
  - It is impossible to keep up alone
  - Shopify uses public slack channels to interact with their coding agent  
<https://x.com/tobi/status/2053121182044451016>
- Vibe coding allows you to build tooling, which would have been impossible before.

# Crafting digital products

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